

Nehal C P

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Publications

[1] **Nehal et al. (2025)**, Investigating the Bulge Morphology of Dual AGN Host Galaxies from the GOTHIC Survey, *MNRAS*, 544, 4208

[2] Bhattacharya et al. (2023), Automated detection of double nuclei galaxies using GOTHIC and the discovery of a large sample of dual AGN, *MNRAS*, 524, 4482 (Co-Author)

[3] Biswas et al. (2026), Investigating the Spectral Properties of Dual Nuclei in Galaxy Mergers from the GOTHIC Survey: Supermassive Black Hole Growth, Metal Enrichment and Dual AGN

Submitted to MNRAS, arXiv: arXiv:2606.20810

Education

BS-MS Dual degree	Indian Institute of Science Education and Research, Bhopal Major: PHYSICS Minor: Electrical Engineering and Computer Science • GPA: 8.14/10 EQF level: 7 Language Medium: English • MS Thesis : DAGN: A study on large sample	July 2018 – July 2023
Higher Secondary Education	St. Michael's Anglo Indian School, Kerala, India • Score: 98.2/100 EQF level: 5 Language Medium: English • Subjects: Physics Chemistry Maths Biology English Hindi	June 2016 – March 2018

Experience

Internship <i>Supervisors:</i> Dr. Mousumi Das, Dr. Sudhanshu Barway (Co-PI) • Analyze the morphological properties of the dual AGN candidate NGC 6240 using JWST/NIRSpec IFU data. • To Perform band-specific feature extraction to probe stellar populations and identify star-forming regions.	IIA Bangalore, India Jan 2026 – March 2026
Internship [Research paper accepted in <i>MNRAS</i> ^[1]] <i>Supervisors:</i> Dr. Mousumi Das, Dr. Sudhanshu Barway (Co. PI) • Python scriptings for SDSS DR18 preprocessing (image extraction etc.) • Performed multi-band (g, r, i) bulge-disk decomposition using GALFIT and iterative refinement. • Estimated bulge stellar masses and merger mass ratios using extinction- and K -corrected colors with standard M/L relations for studies on dual AGN host galaxy to search for trends in bulges and disk distributions.	IIA Bangalore, IN Jan 2024 – May 2025
MS Thesis [Research paper accepted in <i>MNRAS</i> ^[2]] <i>Supervisors:</i> Dr. Mousumi Das (PI), Dr. Rajib Saha (Co-PI) • Retrieved SDSS imaging and spectroscopy data via SQL queries (SkyServer). • Performed data filtering, cleaning, and preprocessing using Python	IIA Bangalore, IN June 2022 – March 2023
Summer Research Fellow, Indian Academy of Science <i>Supervisor:</i> Dr. Mousumi Das • Conducted foundational literature review in observational astronomy.	IIA Bangalore (REMOTE), IN June 2021 – August 2021
Winter Internship , Superconductivity <i>Supervisor:</i> Dr. Ravi Prakash Singh	IISER Bhopal, IN December 2018

Schools / Workshops

Galaxy Mergers and Their Nuclei workshop — Summer School Organized by IIA Bangalore; selected participant from across India	IIA Bangalore, IN 9-11 march 2026
Kodaikanal Solar Observatory — Summer School Organized by IIA Bangalore; selected participant from across India	Kodaikanal, IN June 2024
VIJYOSHI National Science Camp Organized by the Department of Science and Technology (DST), Government of India; for INSPIRE/KVPY scholars across India	IISER Bhopal, IN December 2018

Presentations

Contributed Oral Presentation Galaxy Mergers and Their Nuclei workshop Title: "Bulge morphology studies on Dual AGN systems"	IIA Bangalore, India 9–11 March 2026
VSP seminar Oral presentation Title: "Bulge-disk decomposition of Dual AGN candidates"	IIA Bangalore, IN 24 April 2024
MS Thesis presentation Title: "DAGN: A study on large sample"	IISER Bhopal, IN April 2023
In-House Physics Symposium Poster Presentation	IISER Bhopal, IN 25 March 2023

Coding languages/Softwares

Python | **SQL** (SDSS data extraction) | **GALFIT** (galaxy profile fitting) | **SAOImageDS9** (FITS viewer)
| **LaTeX** | **MATLAB, Origin** (data visualization & analysis) | **COMSOL** (multiphysics simulations) |
TOPCAT (Crossmatching and visualization) |

Research Interests

| High-precision imaging and multi-wavelength studies (IFU) | Galaxy evolution (& SMBH evolution) |
| Astronomical data pipelines architecture and large-scale survey analysis | Instrumentation |
| Time-domain astrophysics and ToO observations (Transient studies) |

Media coverage

- Featured in *Times of India (TOI)*: [New technique helps find thousands of double-nuclei galaxies](#) 
- Featured by *Department of Science and Technology, India*: [Press release on galaxy merger research](#) 
- Highlighted by *DST Official Twitter*: [Research featured on DST social media](#) 